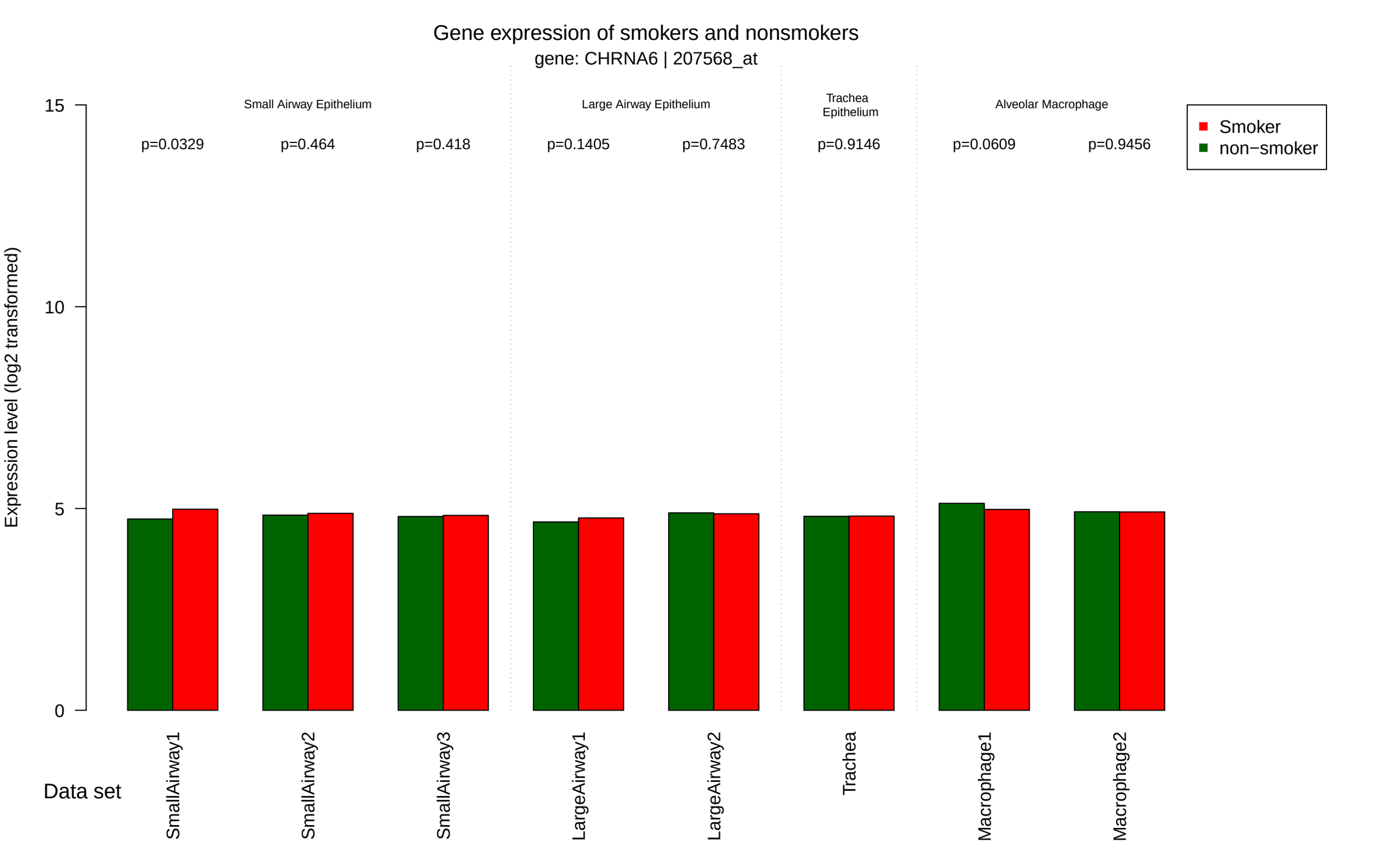
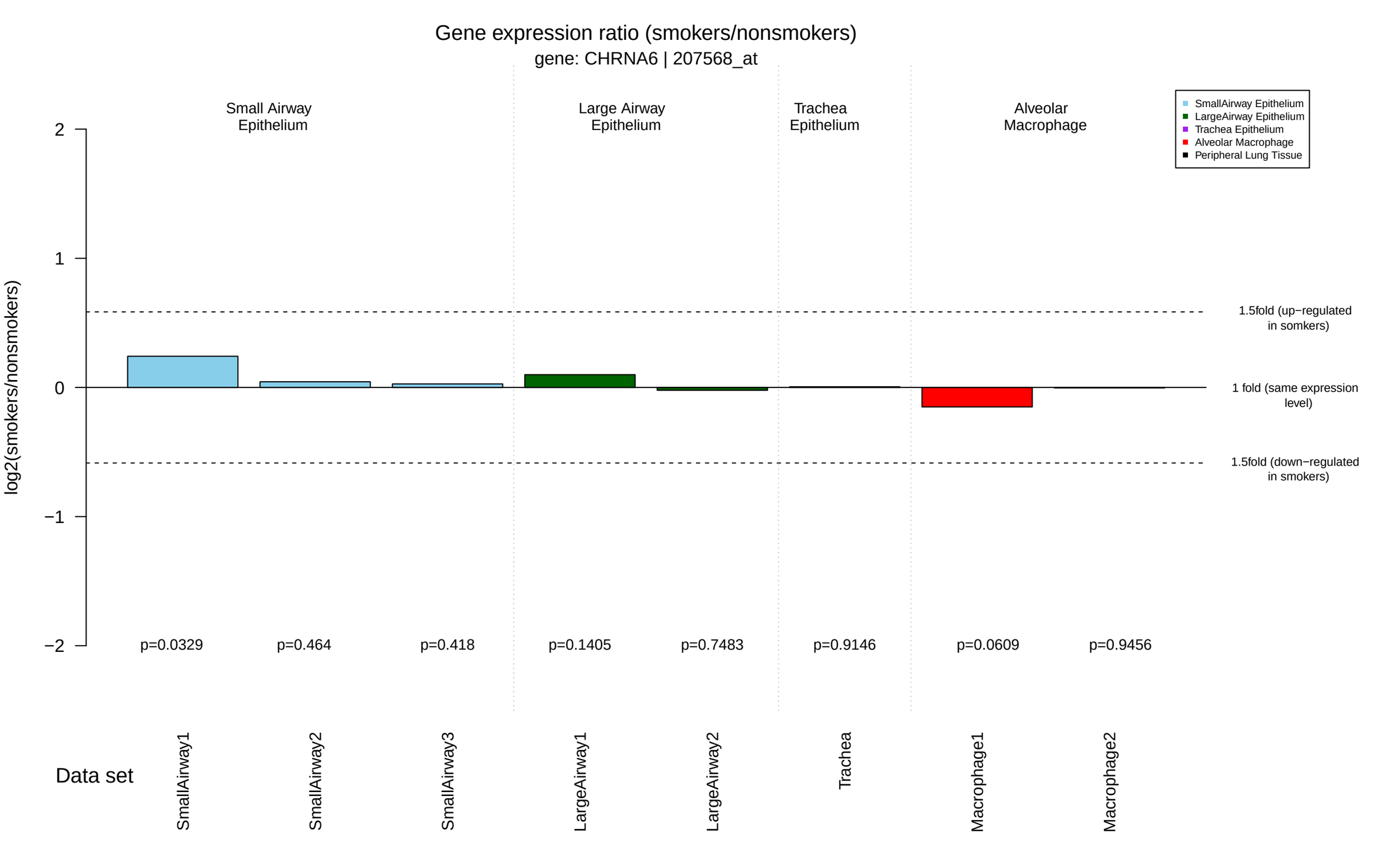
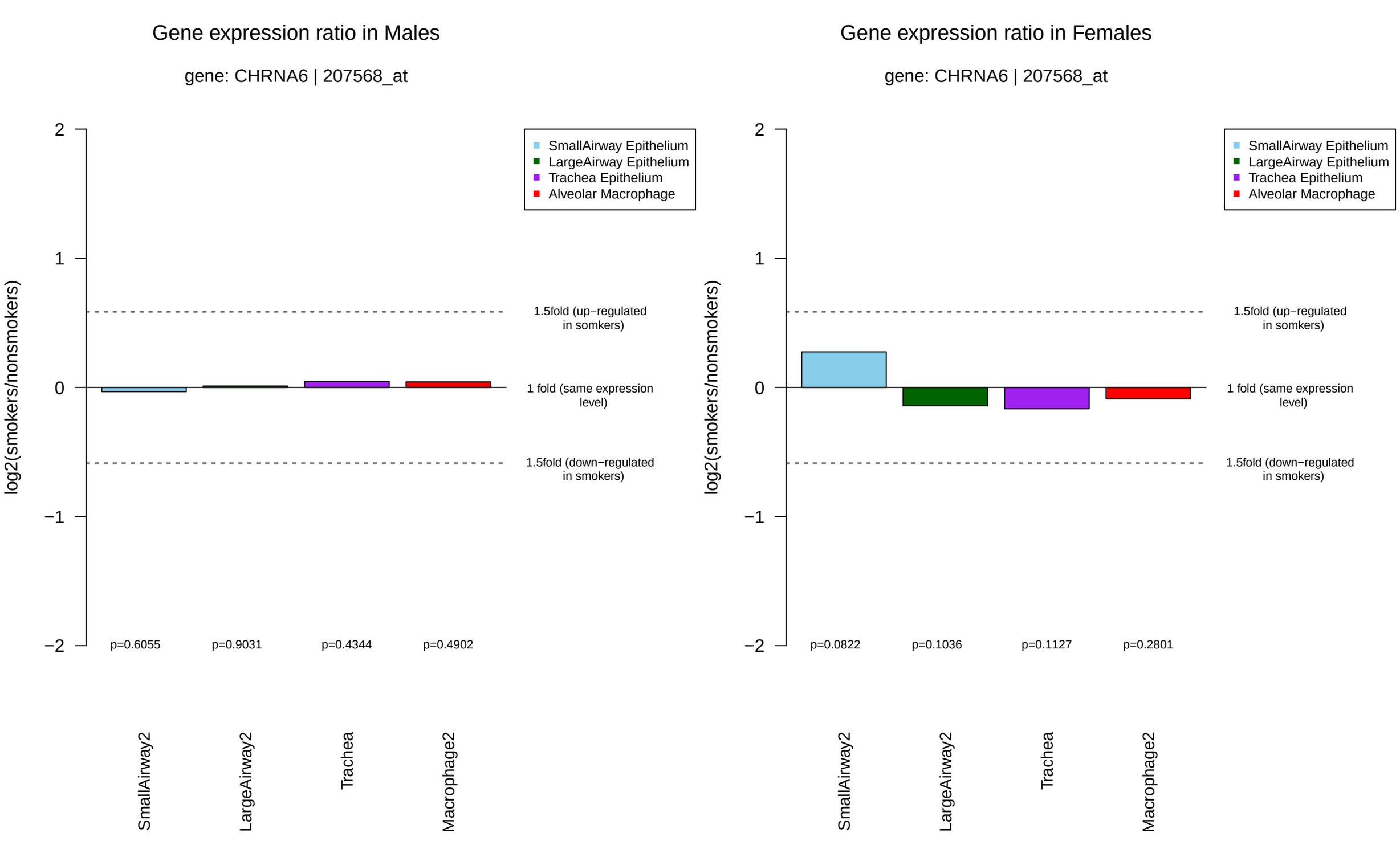
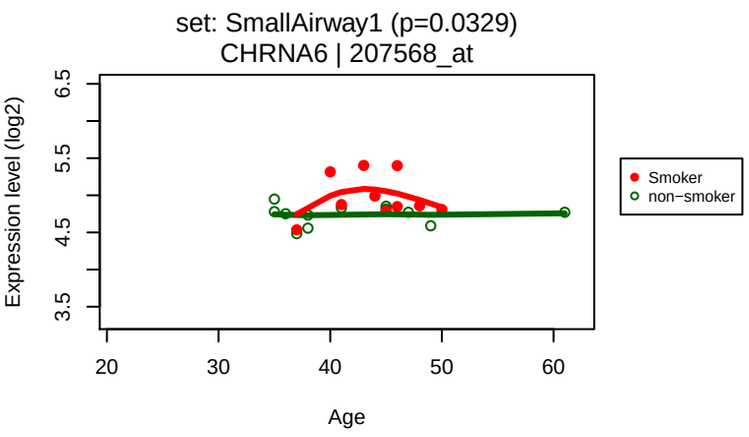
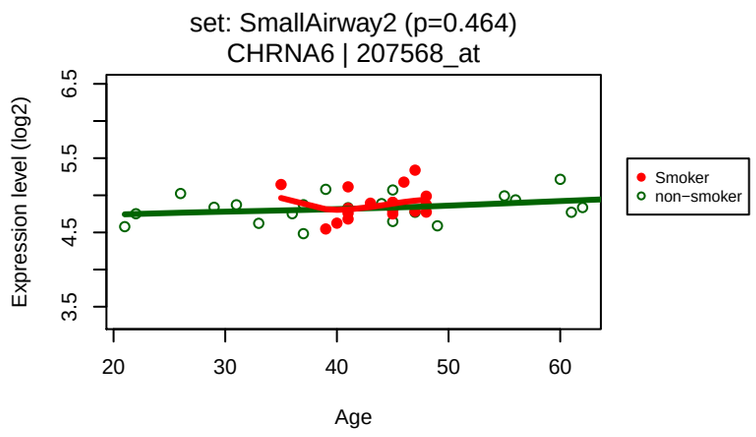
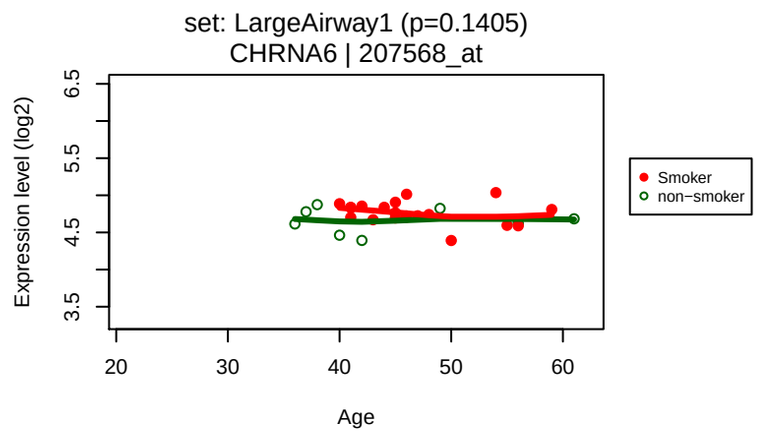

Gene expression of smokers and nonsmokers
gene: CHRNA6 | 207568_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0329
p=0.464
p=0.418
p=0.1405
p=0.7483
p=0.9146
p=0.0609
p=0.9456
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

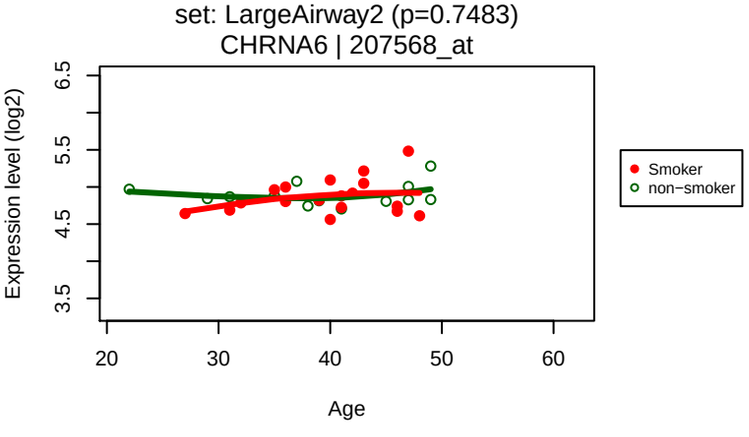

Gene expression ratio (smokers/nonsmokers)
gene: CHRNA6 | 207568_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0329
p=0.464
p=0.418
p=0.1405
p=0.7483
p=0.9146
p=0.0609
p=0.9456
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

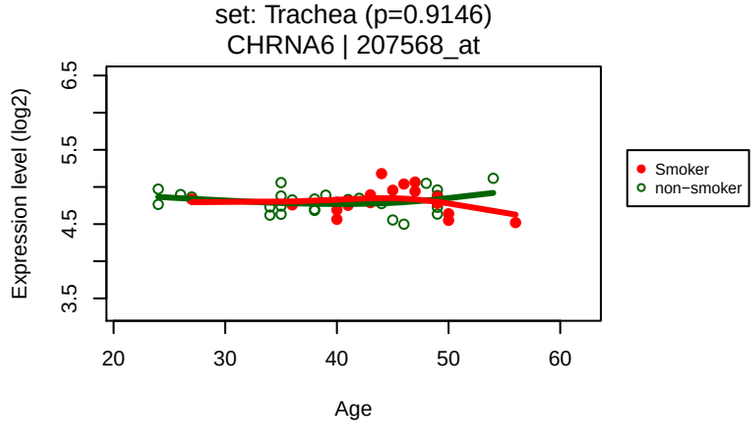

Gene expression ratio in Males
gene: CHRNA6 | 207568_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6055
p=0.9031
p=0.4344
p=0.4902
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHRNA6 | 207568_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0822
p=0.1036
p=0.1127
p=0.2801
SmallAirway2
LargeAirway2
Trachea
Macrophage2

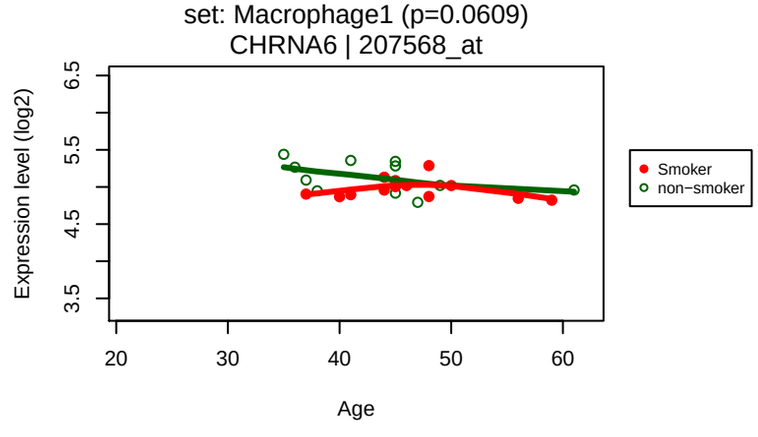

set: SmallAirway1 (p=0.0329)
CHRNA6 | 207568_at
Expression level (log2)
Age
Smoker
non−smoker

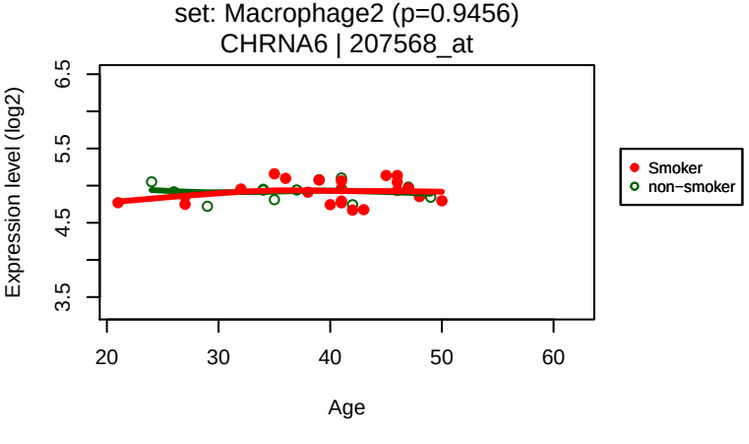

set: SmallAirway2 (p=0.464)
CHRNA6 | 207568_at
Expression level (log2)
Age
Smoker
non−smoker

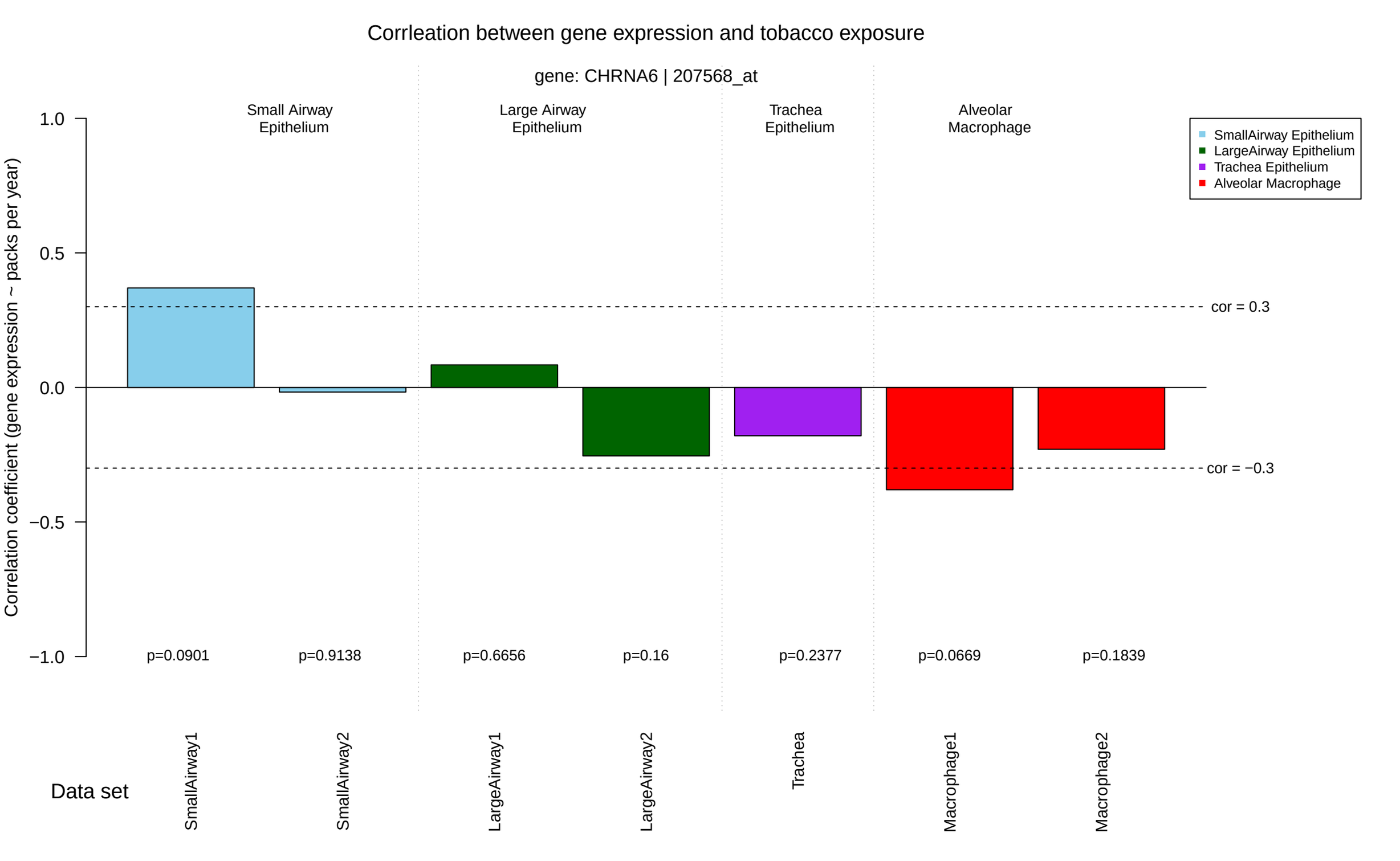

set: LargeAirway1 (p=0.1405)
CHRNA6 | 207568_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7483)
CHRNA6 | 207568_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9146)
CHRNA6 | 207568_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0609)
CHRNA6 | 207568_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9456)
CHRNA6 | 207568_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHRNA6 | 207568_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0901
p=0.9138
p=0.6656
p=0.16
p=0.2377
p=0.0669
p=0.1839
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2